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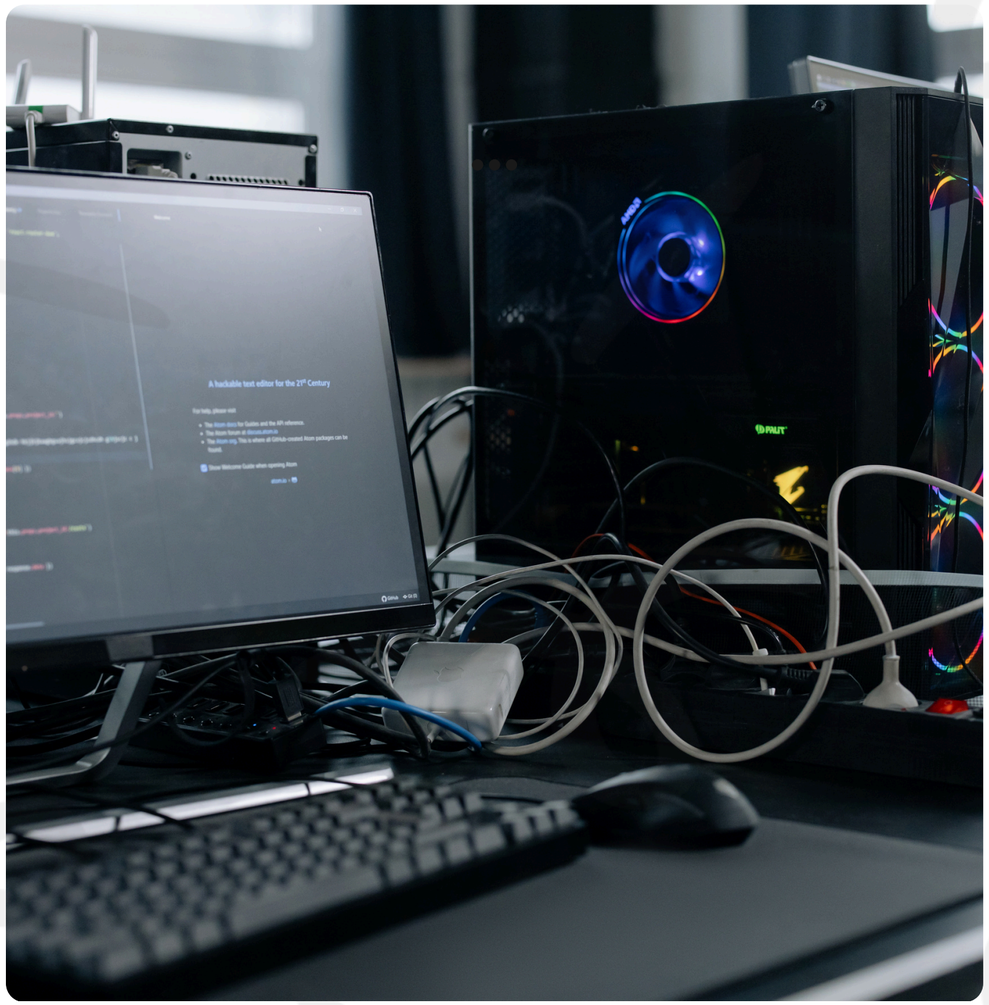
HYBRID QUANTUM-CLASSICAL SENTIMENT ANALYSIS

This project explores a hybrid quantum-classical approach where classical NLP extracts features and a variational quantum circuit performs classification to evaluate potential advantages.



OBJECTIVE

- Build a hybrid quantum-classical sentiment analysis model.
- Convert text features into quantum states using angle encoding through VQC.
- Compare quantum model performance with classical models.
- Demonstrate a working prototype for real-world use.

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IMPLEMENTATION DETAILS

DATASET	News-Headlines dataset (Provides labeled text data for sentiment classification.)
Text Preprocessing	python , NLP (Removes noise and improves feature quality.)
TF–IDF Vectorization	scikit-learn (Converts text into numerical features)
Feature Scaling	MinMaxScaler, PCA (Quantum angle encoding requires values in range [0,π]
Angle Encoding	PennyLane, VQC(Encodes classical data into quantum states.)
Visualization & Demo	Streamlit (Show accuracy graphs, live prediction.)

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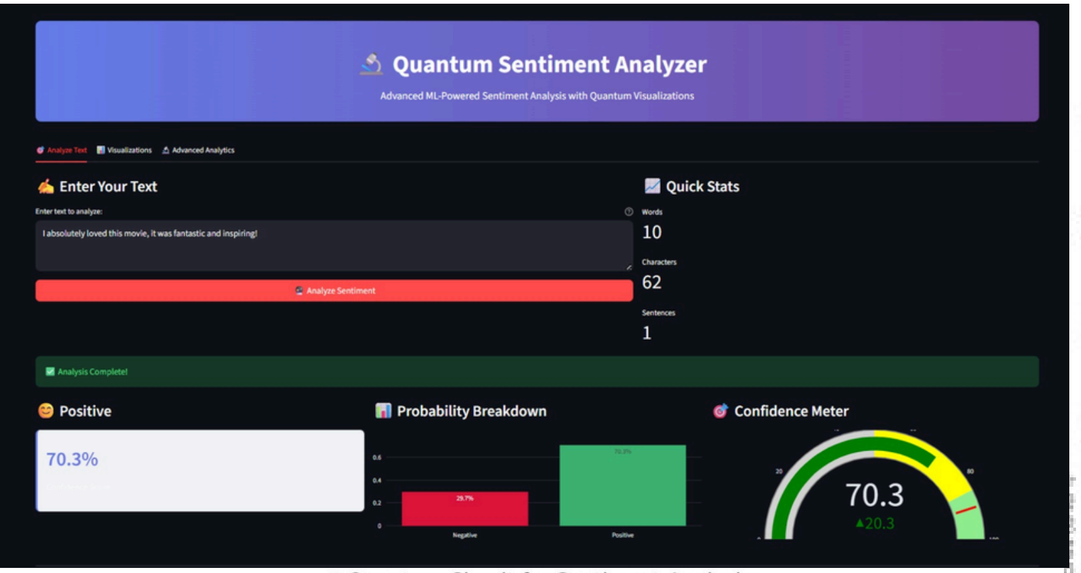
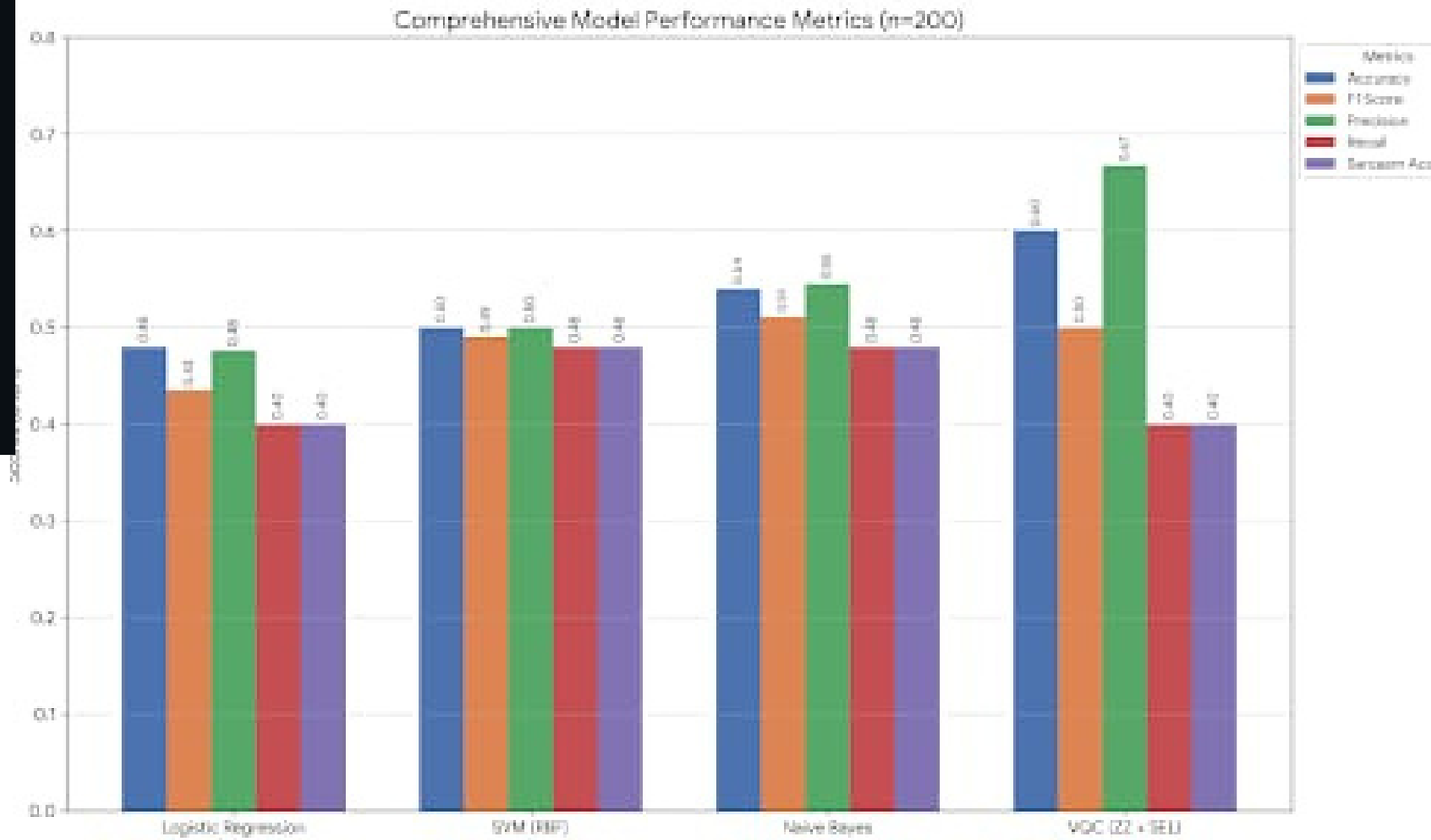
RESULTS



Quantum Circuit for Sentiment Analysis



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Analysis

- The quantum model achieved higher and more stable accuracy than the classical model.
- The hybrid pipeline worked successfully, showing that quantum circuits can be integrated with NLP tasks.

Conclusion

- We built a working hybrid quantum–classical sentiment analysis system.
- The project demonstrates the potential of quantum machine learning for future AI applications.

Future Work

- Improve performance using more qubits and single shot, noise datasets.
- Multimodal, Quantum Attention mechanisms, Explainability.